

Report

Data Mining Project

Table of Contents

1. Introduction
2. Dataset
3. Data Cleaning
4. Algorithms
5. Data Visualization
6. Conclusion

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1. Introduction

This report outlines a data mining project that includes data cleaning, the application of two distinct algorithms, and data visualization techniques in the light of Data Mining & Analytics course curriculum. By applying these methodologies, the project aims to extract meaningful insights from the dataset, facilitate informed decision-making, and drive value for the organization.

2. Dataset:

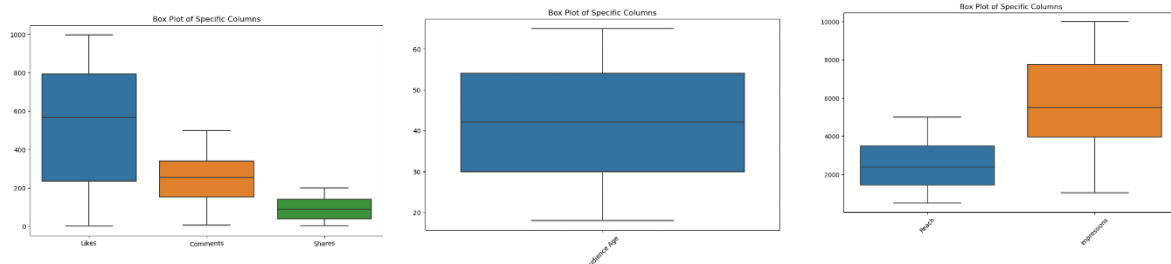
The Social Media Engagement Report dataset is a comprehensive collection of data gathered from various social media platforms over a specific period. It includes metrics related to user interactions, content performance, and audience behavior across different social media channels. The dataset provides insights into the effectiveness of social media marketing efforts, audience engagement trends, and the impact of content strategies on user interactions.

https://www.kaggle.com/datasets/aliredaelblgiyh/social-media-engagement-report?select=social_media_engagement_data.xlsx

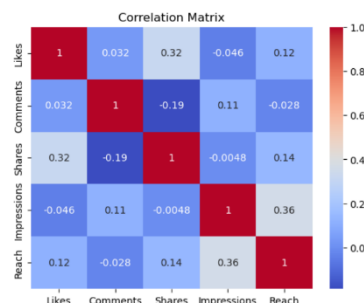
3. Data Cleaning

Data mining involves the extraction of actionable knowledge from raw data. Therefore, data cleaning plays a crucial role in the initial stages of any data mining project. It involves identifying and rectifying errors, inconsistencies, and missing values within the dataset. The dataset utilized in this project was accessed through the following steps:

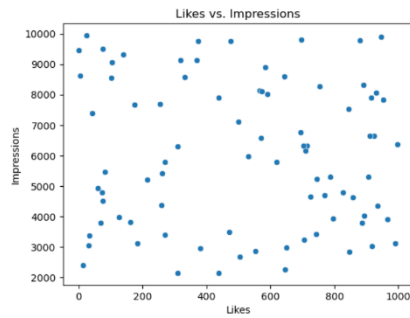
- Removing unnecessary columns, including influencer ID, campaign ID, time, and date
- Handling null values, duplicates, and missing values
- Selected significant columns were visualized using a box plot to identify outliers.
- Data visualization of the attributes to ascertain their distribution and quantity.



- A correlation matrix was generated and printed to examine the relationships between the attributes.



- Generating a scatterplot which allows the examination of the relationship between likes and impressions.

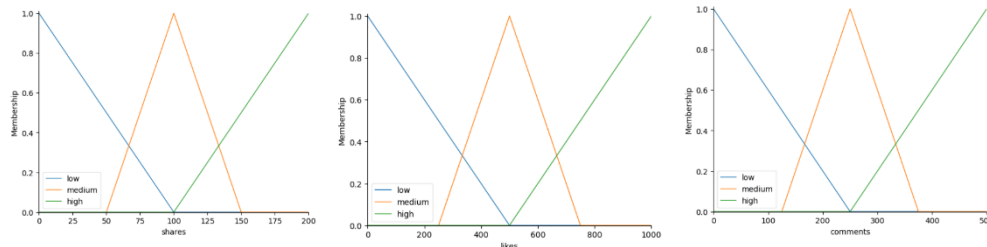


4. Algorithms

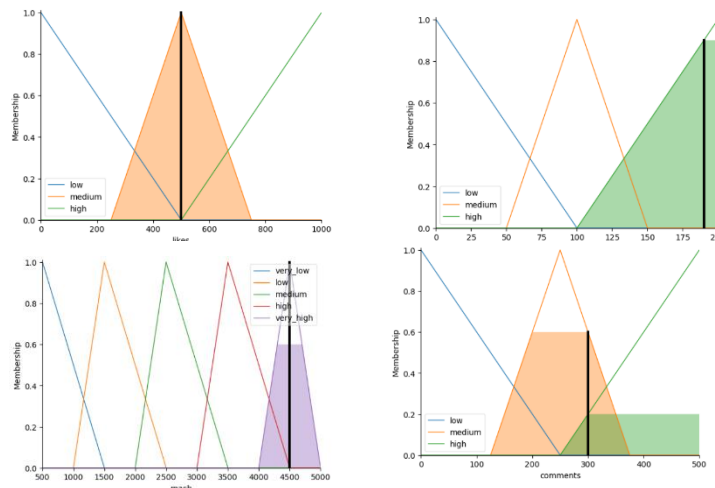
a. Fuzzy Logic Algorithm:

We implemented a fuzzy logic algorithm. Our objective was:

- 1) Identifying the peak values of metrics such as reach, likes, comments, and shares to facilitate the prediction of reach metrics.

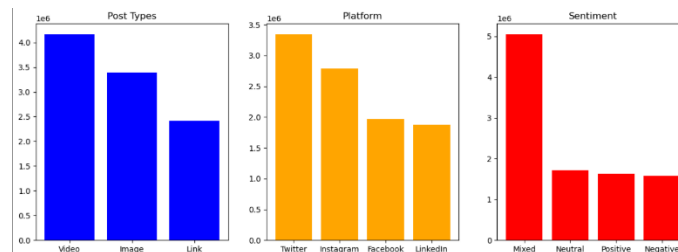


- 2) From the predictions generated by our algorithm, we then categorized the reach into four distinct classes: low, medium, high, and very high.
- 3) To apply fuzzy logic effectively, we defined ranges based on the minimum and maximum values of the metrics involved.
- 4) Acknowledging the attributes and membership functions, we specified 27 rules.
- 5) Following the establishment of 27 rules, we applied them to a sample of 100, expanding the dataset with two additional columns: “predicted reach” and “reach classification”.
- 6) Finally, visualization the fuzzy membership function

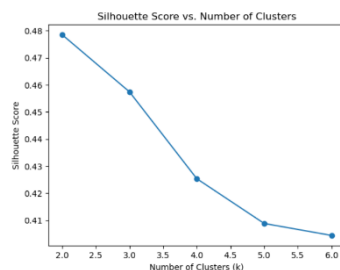


b. *K-medoids Algorithm*

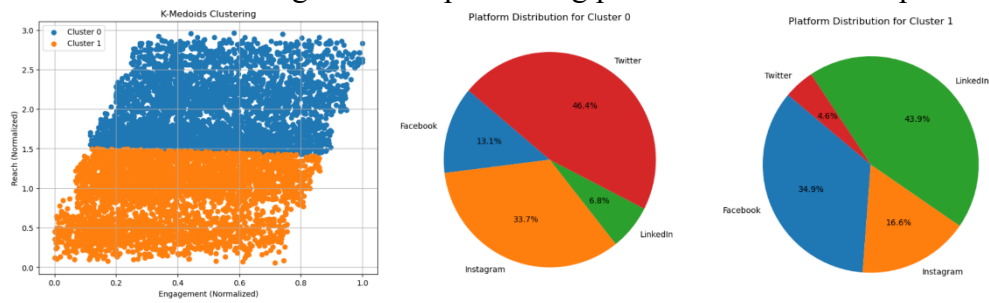
- 1) We implemented a fuzzy logic algorithm. Our objective was:
 - Identify which combinations of platform and post type typically yield superior performance.
 - Evaluate performance based on two key metrics: Engagement (likes, comments, and shares), reach.
- 2) Through visualization, we examine how likes are distributed across various post types, platforms, and sentiment categories.



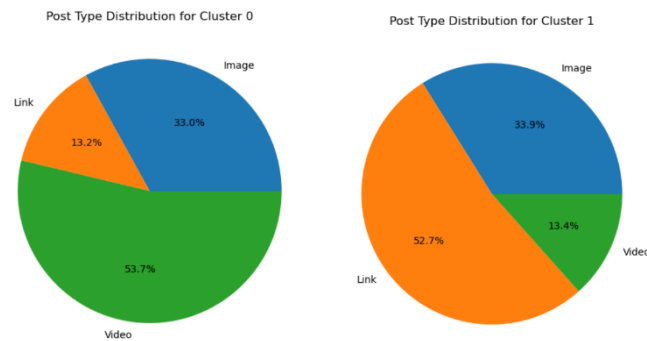
- 3) In summary, the silhouette score provides a way to evaluate the quality of clustering results by considering both how close the points are within clusters and how well-separated the clusters are from each other. Higher silhouette scores indicate better clustering.
- 4) We will execute K-medoids clustering and assess silhouette scores to ascertain the optimal number of clusters. Silhouette scores offer a means to gauge the clustering quality by assessing the proximity of points within clusters and the separation between clusters. Elevated silhouette scores signify superior clustering outcomes.



- 5) Selecting the cluster exhibiting higher engagement metrics, followed by determining the better performing platforms within this superior cluster.



- 6) After identifying that Twitter and Instagram typically yield better performance, we then analyze which post type performs better among video, image, and link posts.



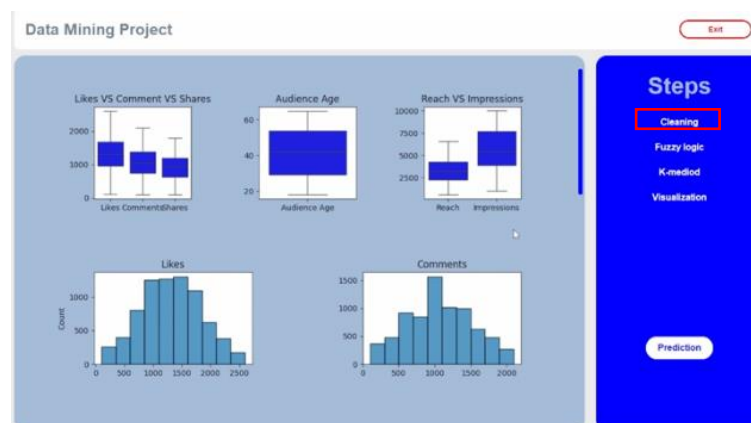
- 7) We conclude that videos perform better.

5. Visualization

Using visualizations and a GUI users can quickly understand key insights and make informed decisions. The GUI enables users to dynamically adjust parameters, perform iterative analysis, facilitate clearer communication, and enhance collaboration among stakeholders.

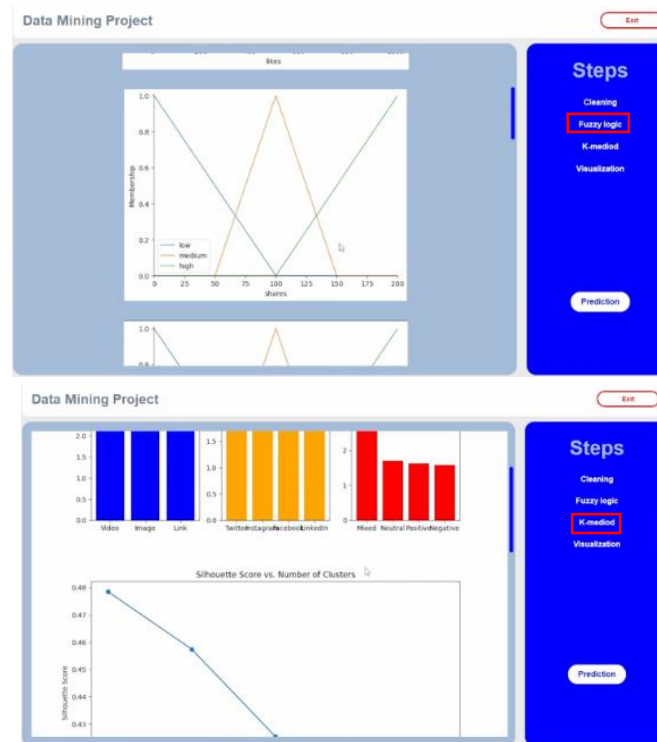
- *Cleaning:*

Using a GUI and visualizations makes data cleaning easier, lets users set tasks and parameters, while visualizations help spot errors and patterns for quicker corrections. This approach boosts productivity and ensures data quality.



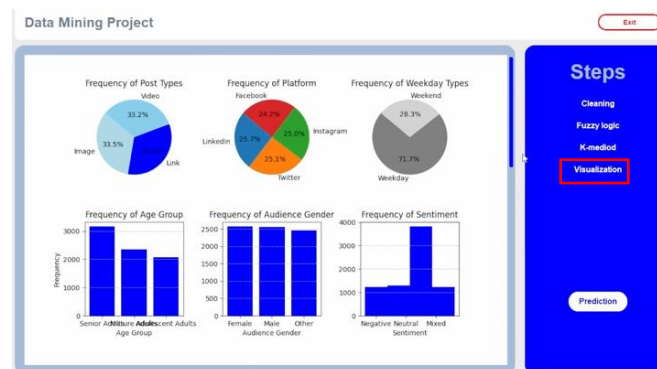
- **Fuzzy Logic and K-Medoids:**

The GUI provides an intuitive platform for users to interact with the algorithms, visualize intermediate results, and interpret the final outcomes. Visualizations help users understand the algorithm's behavior, assess its performance, make informed decisions, explore different settings and compare results.



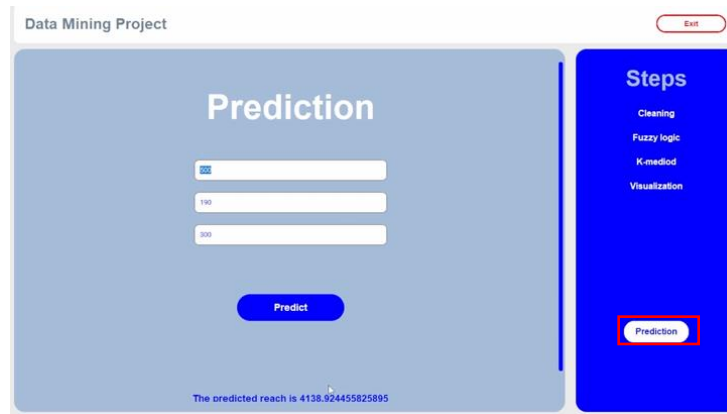
- **Visualization:**

Visualizing the frequency distributions of platform, post types, weekday types, age groups, sentiment, and audience gender provides valuable insights into the patterns within the dataset. By creating histograms and pie charts for each variable, users can easily observe the distribution of data across different categories. These visualizations help identify dominant categories, detect any imbalances or biases in the dataset, and uncover potential relationships or trends between variables.



- **Prediction:**

Reach prediction simplifies the process, allowing users to input engagement metrics. Users can explore various scenarios fostering better decision-making. The GUI's user-friendly interface enables customization and error checking, ensuring accurate predictions.



The screenshot displays a web application titled "Data Mining Project". The main area is a light blue box labeled "Prediction" containing three input fields with values "25", "100", and "200", and a blue "Predict" button. Below the button, it states "The predicted reach is 4138.924455825895". On the right, a dark blue sidebar titled "Steps" lists "Cleaning", "Fuzzy logic", "K-mediod", and "Visualization", with a "Prediction" button at the bottom highlighted by a red rectangle. An "Exit" button is in the top right corner.

6. Conclusion:

In summary, this data mining project utilized data cleaning, fuzzy logic, K-medoids clustering, and visualization techniques to extract insights from the Social Media Engagement Report dataset. Through meticulous cleaning and analysis, we gained valuable insights into platform performance, post types, audience demographics, and sentiment analysis. The incorporation of a user-friendly GUI facilitated exploration and decision-making, resulting in actionable intelligence to optimize social media strategies.